

IOWA STATE UNIVERSITY

Department of Computer Science

‘Deep Learning-Based Intelligent Surveillance System for Theft Detection and Animal Behavioral Analytics.

COURSE - HCI-5750- COMPUTATIONAL PERCEPTION

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Background

- AI and computer vision are transforming modern surveillance systems.
- Farms, warehouses, and wildlife areas require **continuous monitoring**.
- Theft incidents are increasing in agricultural and rural environments.
- Monitoring animal behavior is important for **livestock health and safety**.
- Traditional surveillance systems only detect motion and lack intelligent analysis.

Goal:

Develop an intelligent system that can detect humans and monitor animal behavior simultaneously.

Problem Statement

Existing surveillance systems have several limitations:

- Only detect **motion**, not object type.
- Cannot distinguish **human vs animal activities**.
- Limited ability to analyze **behavior patterns**.
- No integration of **security monitoring and animal welfare monitoring**.

Solution:

Develop an AI-based system capable of **real-time human detection, animal monitoring, and behavior classification**.

Proposed System Architecture

Video Input

- Capture real-time video from surveillance cameras

Object Detection

- Detect humans and animals

Tracking

- Track movement across frames

Behavior Analysis

- Analyze movement patterns and inactivity

Alert System

- Generate alerts for suspicious or abnormal activity

Theft Detection

The theft detection module identifies suspicious human activities.

Method Used

- Histogram of Oriented Gradients (**HOG**)
- Support Vector Machine (**SVM**) classifier

Process

- Detect humans in video frames
- Draw bounding boxes around detected persons
- Analyze behavior patterns

Suspicious Activities

- Loitering near restricted areas
- Unauthorized entry
- Repeated movement in sensitive zones

Alerts are generated when suspicious behavior is detected.

Animal Detection and Monitoring

The system also monitors animal movement and behavior.

Technique Used

- Background subtraction
- Contour detection

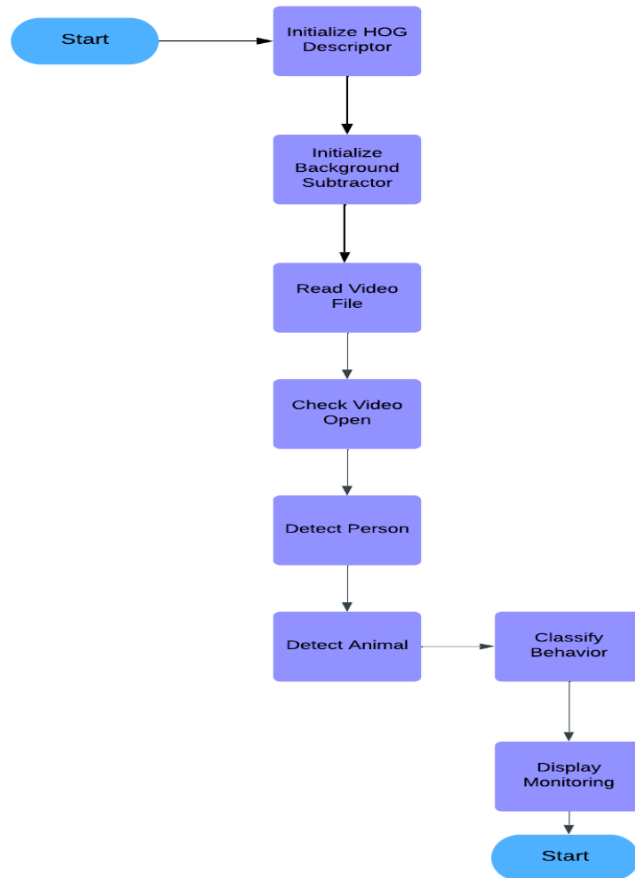
Key Steps

- Detect animal movements in video frames
- Track animal positions using centroid tracking
- Monitor movement patterns

Purpose

- Ensure animal health and safety
- Identify abnormal behavior early

Algorithm Flowchart



Algorithm	Description	Advantages	Disadvantages
HOG (Histogram of Oriented Gradients)	An image descriptor that extracts features based on gradient orientation to detect objects like people.	- Effective for object detection tasks - Robust against illumination changes - Good performance in real-time applications	- Sensitive to variations in scale and pose - May require extensive training data for high accuracy
Background Subtraction (MOG2)	A technique for separating foreground objects (like animals) from the background in video streams.	- Adapts to changes in lighting conditions - Effective in dynamic environments	- May struggle with shadows and reflections - Requires careful tuning of parameters
Speed Calculation	Calculates the speed of moving objects (animals) based on their position in consecutive frames.	- Provides insights into behavior - Useful for detecting abnormal behaviors based on speed	- Requires accurate tracking of positions - Sensitive to frame rate and motion blur
Behavior Classification	Classifies animal behavior based on inactivity duration and movement patterns.	- Identifies abnormal behaviors - Simple logic for interpretation	- Reliant on thresholds which may need adjustments - Can produce false positives or negatives
Animal Classification	Classifies animals based on the size of the detected bounding box in the frame.	- Simple and fast classification - Effective in controlled environments	- Limited to size-based classification - May misclassify with overlapping objects

Behavior Classification

Animal behavior is classified based on movement patterns.

Parameters Used

- Speed of movement
- Inactivity duration
- Erratic movement patterns

Behavior Categories

- Normal behavior
- Slow movement
- Prolonged inactivity
- Erratic movement

Abnormal behaviors trigger alerts for further investigation.

Implementation

The system is implemented using modern computer vision tools.

Programming Language

- Python

Libraries and Tools

- OpenCV (video processing)
- Machine learning algorithms
- Background subtraction techniques

Processing Steps

- Capture video frames
- Detect humans and animals
- Track movements
- Analyze behavior patterns

The system continuously processes video streams in real time.

Result Analysis

The system was tested on surveillance video data.

Person Detection

- Humans detected using HOG descriptor
- Bounding boxes highlight detected persons
- Alert sound generated when person detected

Animal Monitoring

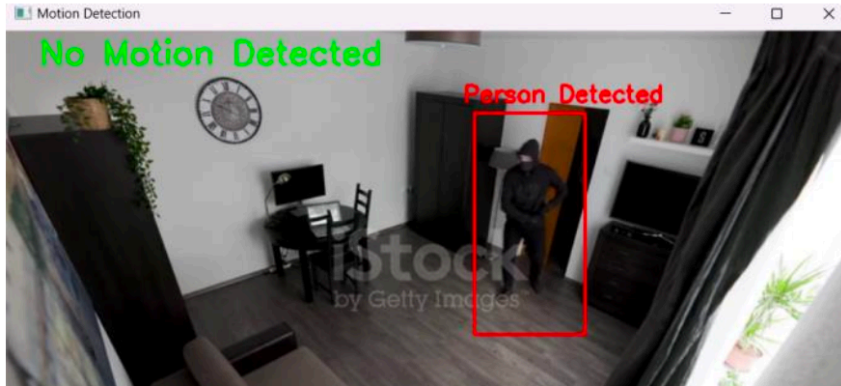
- Detects movement patterns of animals
- Identifies abnormal behaviors

Speed Analysis

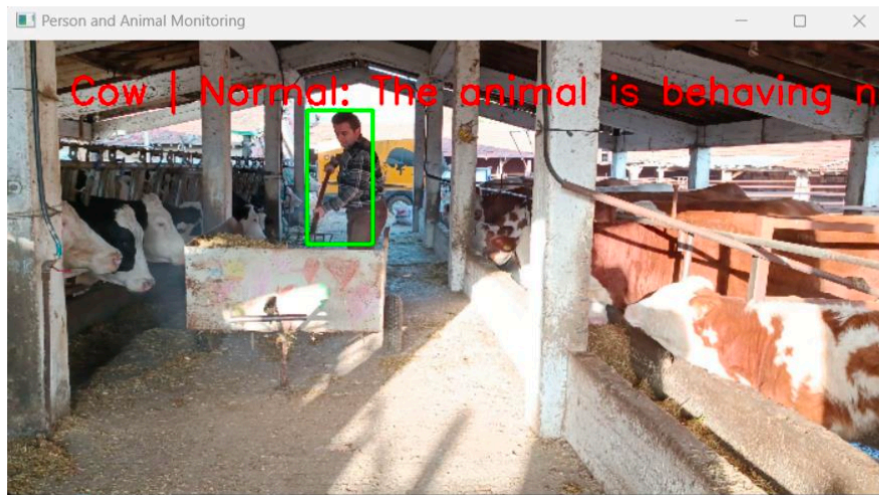
- Calculates animal speed using centroid tracking

Overall, the system successfully detects human activity and monitors animal behavior simultaneously.

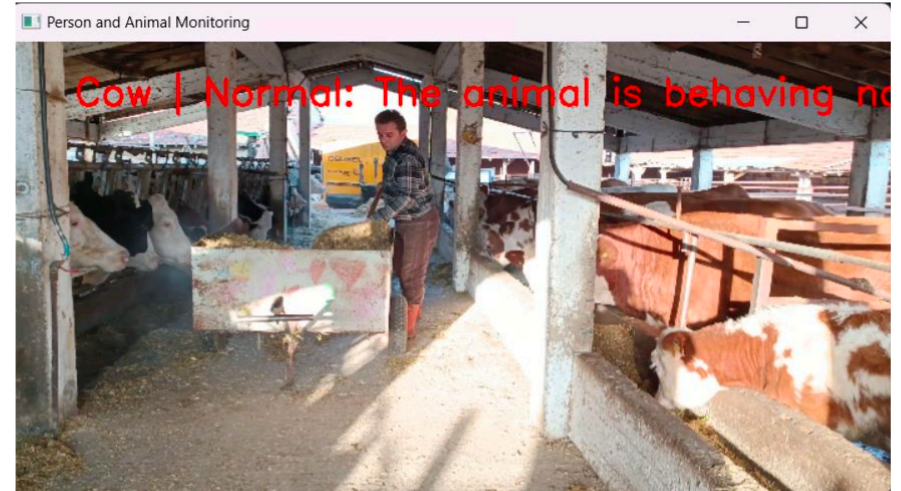
Results for example



Output Screenshot



Output:



Theft Detection

Animal Monitoring



Thank You

Feel free to ask any questions.